

SOLUTION

QUESTION 1

Aim

To implement the Resolution Algorithm and prove that the ground is wet from the given knowledge base.

Propositional Representation

- **R** = It is raining.
- **W** = Ground is wet.

Rule

If it rains, then the ground becomes wet.

$$\mathbf{R \rightarrow W}$$

Fact

R

Goal

W

CNF Conversion

Using:

$$\mathbf{P \rightarrow Q \equiv \neg P \vee Q}$$

Therefore:

$$\mathbf{R \rightarrow W \equiv \neg R \vee W}$$

The CNF clauses are:

- **C1 = $\neg R \vee W$**
- **C2 = R**

Negation of Goal

Goal:

W

Negated goal:

$\neg W$

Therefore:

- **$C3 = \neg W$**

Resolution Steps

Step 1

Resolve:

C1: $\neg R \vee W$

with:

C2: R

Resolving **$\neg R$** and **R** gives:

C4: W

Step 2

Resolve:

C4: W

with:

C3: $\neg W$

Resolving **W** and **$\neg W$** gives:

□

The empty clause is obtained.

Result

Therefore, the goal **W** is logically proved.

Conclusion: The ground is wet.

QUESTION 2

Aim

To implement the Resolution Algorithm and prove that Rahul receives internal marks after submitting his assignment.

Propositional Representation

- S = Rahul submitted the assignment.
- M = Rahul receives internal marks.

Rule

$$S \rightarrow M$$

Fact

$$S$$

Goal

$$M$$

CNF Conversion

Using:

$$S \rightarrow M \equiv \neg S \vee M$$

Therefore:

- $C1 = \neg S \vee M$
- $C2 = S$

Negation of Goal

Goal:

$$M$$

Negated goal:

$$\neg M$$

Therefore:

- $C3 = \neg M$

Resolution Steps

Step 1

Resolve **C1** and **C2**:

$(\neg S \vee M)$

S

Therefore:

C4 = M

Step 2

Resolve **C4** and **C3**:

M

$\neg M$

Therefore:

□

Result

The empty clause is obtained.

QUESTION 3

Aim

To implement the Resolution Algorithm and prove that Priya can borrow books.

Propositional Representation

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Rule

$L \rightarrow B$

Fact

L

Goal

B

CNF Conversion

Using:

$L \rightarrow B \equiv \neg L \vee B$

Therefore:

- **$C1 = \neg L \vee B$**
- **$C2 = L$**

Negation of Goal

Goal:

B

Negated goal:

$\neg B$

Therefore:

- **$C3 = \neg B$**

Resolution Steps

Step 1

Resolve:

$C1: \neg L \vee B$

with:

$C2: L$

Therefore:

C4 = B

Step 2

Resolve:

C4: B

with:

C3: $\neg B$

Therefore:

□

Result

The empty clause is obtained.

Conclusion: Priya can borrow books is logically proved.

QUESTION 4

Aim

To implement the Resolution Algorithm and prove that Arun is eligible for placement.

Propositional Representation

- **A** = Arun cleared the aptitude test.
- **P** = Arun is eligible for placement.

Rule

$A \rightarrow P$

Fact

A

Goal

P

CNF Conversion

Using:

$$\mathbf{A \rightarrow P \equiv \neg A \vee P}$$

Therefore:

- $\mathbf{C1 = \neg A \vee P}$
- $\mathbf{C2 = A}$

Negation of Goal

Goal:

P

Negated goal:

$\neg\mathbf{P}$

Therefore:

- $\mathbf{C3 = \neg P}$

Resolution Steps

Step 1

Resolve:

$$\mathbf{C1: \neg A \vee P}$$

with:

$$\mathbf{C2: A}$$

Therefore:

$$\mathbf{C4 = P}$$

Step 2

Resolve:

C4: P

with:

C3: $\neg P$

Therefore:

□

Result

The empty clause is obtained.

Conclusion: Arun is eligible for placement is logically proved.

QUESTION 5

Aim

To implement the Resolution Algorithm and prove that the user is granted access.

Propositional Representation

- **P** = User entered the correct password.
- **A** = User is authenticated.
- **G** = User is granted access.

Rule 1

$P \rightarrow A$

Rule 2

$A \rightarrow G$

Fact

P

Goal

G

CNF Conversion

Rule 1

$$P \rightarrow A$$

Therefore:

$$C1 = \neg P \vee A$$

Rule 2

$$A \rightarrow G$$

Therefore:

$$C2 = \neg A \vee G$$

Fact

$$C3 = P$$

Negation of Goal

Goal:

$$G$$

Negated goal:

$$\neg G$$

Therefore:

$$C4 = \neg G$$

Resolution Steps

Step 1

Resolve C1 and C3:

$$\neg P \vee A$$

$$P$$

Therefore:

$$C5 = A$$

Step 2

Resolve C2 and C5:

$\neg A \vee G$

A

Therefore:

$C6 = G$

Step 3

Resolve C6 and C4:

G

$\neg G$

Therefore:

□

Resolution Process

Correct Password

↓

Authenticated

↓

Granted Access

↓

$\neg$ Granted Access

↓

□ **Empty Clause**

Result

The empty clause is obtained.

Conclusion: The user is granted access is logically proved using Resolution Refutation.

